NAME: ____________________________________ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

MARKING SCHEME

BIOLOGY
Paper 4 Practical

9744/04
26 August 2021

2 hours 30 minutes

Candidates answer on the Question Paper

Additional Materials:

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
TOTAL	

Answer **all** questions

Before you proceed, read carefully through **the whole** of Questions 1 and 2.

Plan the use of your **time** to make sure that you finish all the work that you would like to do.

You will **gain marks** for recording your results according to the instructions.

1 During the light dependent stage of photosynthesis, hydrogen ions and electrons are transferred to hydrogen acceptor molecules, including NADP.

DCPIP (2,6-dichlorophenolindophenol) is a blue dye, which acts as a hydrogen ion and electron acceptor. As DCPIP accepts hydrogen ions or electrons it is reduced and becomes colourless.

You are required to investigate the effect of different colours of light on the rate of the light dependent stage of photosynthesis in a leaf extract containing chloroplasts.

You are provided with:

- a leaf extract in buffered solution, labelled **L**, in a beaker with ice,
- DCPIP solution, labelled **D**,
- filters that allow light of specific colours to pass through, as shown in Table 1.1.

Table 1.1

colour filter	label
purple	**U**
green	**G**
orange	**O**
red	**R**

The leaf extract is an irritant. It is recommended that you wear safety goggles and gloves.

Proceed as follows.

1 Stir the leaf extract, **L**, using the glass rod.

2 Use a syringe to draw up 0.5 cm^3 of **L**.

3 Wipe the outside of the syringe to remove any liquid.

4 Put the syringe at the centre of a white tile. This will be used as a colour standard.

5 You are required to add 0.2 cm^3 DCPIP solution, **D**, to change the colour of the remaining leaf extract, **L**. The change in colour must be sufficient to be observable in the 0.5 cm^3 sample transferred to a syringe in step **8**. If the change in colour is not observable, add one or two more drops of DCPIP solution. You are advised not to add more than 0.5 cm^3 of DCPIP solution in total.

6 Immediately wrap the specimen tube containing the mixture of **L** and **D** in foil. Cover the specimen tube with a foil lid, as shown in Fig. 1.1. This should be easy to remove to obtain the mixture of **L** and **D**. Put back the covered specimen tube containing the mixture of **L** and **D** in the styrofoam cup with ice water.

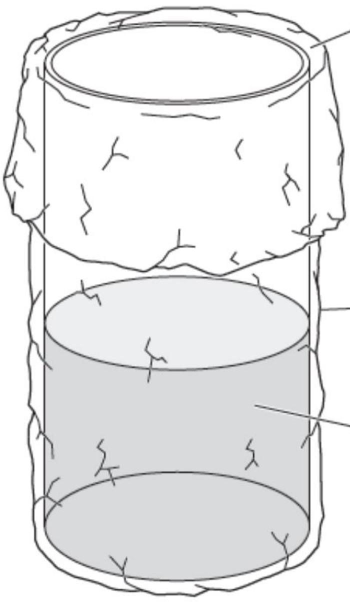


Fig. 1.1

7 Place the bench lamp at about 10 cm from the syringe on the white tile. Do **not** switch the lamp on.

The next steps have to be carried out very quickly one after another, so read steps 8 to 15 and refer to Fig. 1.2 before proceeding.

8 Remove the foil lid and use a clean syringe to draw up 0.5 cm^3 of the mixture of **L** and **D** in the specimen tube. Replace the foil lid immediately.

9 Wipe the outside of the syringe and place it next to the colour standard on the white tile. This is the test syringe.

10 Immediately cover both syringes with the purple filter, **U**, as shown in Fig. 1.2.

11 Switch on the bench lamp and immediately start a stopwatch.

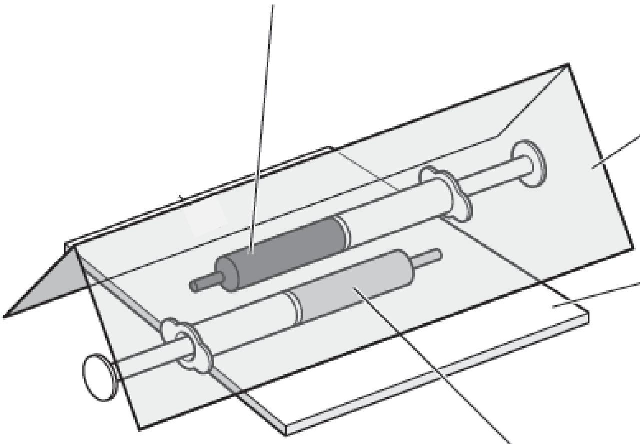


Fig. 1.2

12 Measure the time taken for the colour in the test syringe to match that of the colour standard. If the colour does not match after 300 seconds, then record 'more than 300'. Record your results in **(a)(i)**.

13 Switch off the bench lamp.

14 Expel the contents of the test syringe into the beaker labelled **waste**. Rinse the syringe.

15 Repeat steps 8 to 14 using each of the three remaining colour filters (**G**, **O**, and **R**) in turn.

(a) **(i)** Use the space to record, in a suitable format, your results from step **12**.

Coloured filter	**Time to decolourise DCPIP / s**
U / purple	130 - 166
G / green	more than 300
O / orange	170 – 263
R / red	214 - 243

[H]: Suitable headings with appropriate units; leftmost column for independent variable

[O]: Correct recording of time to decolourise DCPIP; precision up to whole number

[T]: Correct trend; U: the fastest (shortest time); followed by O/R; and lastly green (has to be more than 300);

[3]

(ii) Explain why the leaf extract was kept in ice water.

....

.... [1]

1. (Idea of) **stopping all enzymatic activity in the leaf extract** and preventing the enzymes from damaging the chloroplasts

Reject: to slow down photosynthesis or vague answers pertaining to slowing down reactions.

(iii) Explain why the leaf extract containing DCPIP was kept covered by foil.

....

.... [1]

1. (Idea of) otherwise photosynthesis would occur due to the incident / ambient / surrounding light source and decolourise the DCPIP

(iv) Describe a suitable control that could have been set up for this investigation. Justify your answer.

....

....

....

.... [2]

1. Using a clean syringe, draw up **<u>0.5 cm^3</u>** of a mixture containing **DCPIP and water**;
2. to show the **change in colour of the DCPIP over time** is due to the **presence of the leaf extract**.

OR

3. Using a clean syringe, draw up **<u>0.5 cm^3</u>** of a mixture containing **DCPIP and boiled and cooled leaf extract**;
4. to show the **change in colour of the DCPIP over time** is due to the **presence of the active cellular components in the leaf extract**.

OR

5. Using a clean syringe, draw up **<u>0.5 cm^3</u>** of a mixture containing **DCPIP and leaf extract and kept in the dark**;
6. to show the **change in colour of the DCPIP over time** is due to the **effect of light on the leaf extract**.

OR

7. Using a clean syringe, draw up 0.5 cm3 of **boiled and cooled leaf extract only**;
8. to show that the **change in colour in the other syringes** is due to the **presence of DCPIP**.

(v) Explain why rate of photosynthesis is different with the different colour filters.

....

....

....

.... [2]

1. Some wavelengths that passes through the colour filters are more effective than others for photosynthesis, e.g. purple / red / orange coloured filters allow wavelengths of light that are more effective for photosynthesis compared to green filters.

2. Ref to. plant chlorophyll which reflect green wavelength while red / purple / orange wavelengths of light had higher absorbance by chlorophyll.

(vi) The results of this investigation may be limited by lack of replication and sources of error.

Describe **one other** significant source of error in the procedure and suggest an improvement that can be made.

....

....

....

.... [2]

1. Difficulty in matching the colour of the standard syringe by using **visual inspection** (or by eye) which can be subjective.
2. Use a **colourimeter**, calibrated and zeroed with the content as the colour standard syringe, to monitor the change in colour of the contents from the other syringes.

OR

3. Inability to see the colour change all the time having to keep the filter in place.
4. Use of **colour sensors within the syringes**, linked to an external moitor, to allow constant comparison of colours without the need to lift the filter.

OR

5. Actual wavelength of light used affected by leakage of ambient light / white light from bulb through the ends of the folded filter.
6. **Remove all other light sources** / use **light source** from a **narrow light beam** / filtered through a narrow slit.

OR

7. Warming up of the contents in the syringes, hence affecting colour change in DCPIP, due to the heating effect of the lamp.

   [Reject: due to changes in temperature as a result of fluctuations in the ambient temperature – since this would affect the contents in both the test syringes and the standard syringes equally.]

8. Use of a **heat filter, placed in front of the bulb** OR use a **cool light source**.

(b) In another similar investigation, a student collected leaves from two varieties of the same species of a garden plant that has different coloured leaves.

Variety **A**	dark red leaves
Variety **B**	green and white striped leaves

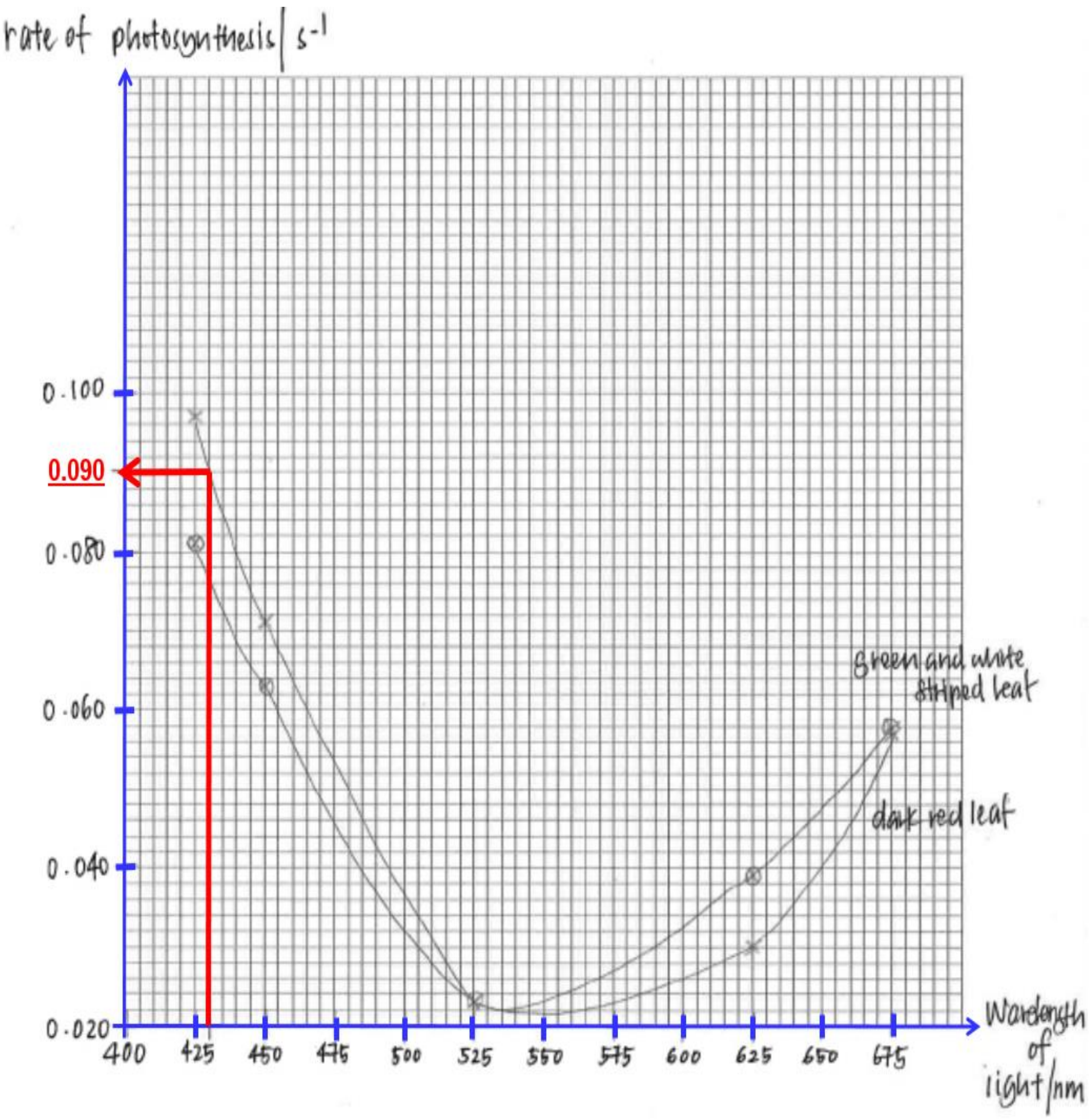
The student made a chloroplast extract from the leaves of each variety and measured the rate of photosynthesis for each extract in different wavelengths of light.

Table 1.2 shows the rates of photosynthesis calculated by the student from her results.

Table 1.2

wavelength of light / nm	rate of photosynthesis / s^{-1}	
	dark red leaf	green and white striped leaf
425	0.097	0.081
450	0.071	0.063
525	0.023	0.023
625	0.030	0.039
675	0.057	0.058

(i) Use the grid provided to display the results shown in Table 1.2 in appropriate form.



Mark Allocation:

[S] – **appropriate Scale** used for both axes, with **equidistant intervals / equally-spaced intervals;** graph to occupy at least **more than half of the given grid** (in both the x and y directions)

[A] – for **correctly labelled Axes with correct units** [rate of photosynthesis / s^{-1}] and [wavelength of light / nm]

[P] – **all Points accurately plotted** (within half a square)

[L] – for **use of straight point-to-point Line** to join all plot points [see above] **or smooth curve (i.e. line of best fit)** [see below] **AND NO EXTRAPOLATION beyond extreme plotted points**

[4]

(ii) Use the graph in **(i)** to estimate the rate of photosynthesis at a wavelength of 430 nm in plants with dark red leaves.

.... [1]

Accept: 0.090 s^{-1} to 0.095 s^{-1};

*[Reject: if student did not show how he/she estimated the rate of photosynthesis using the graph in **(i)]**.*

(c) A student deduced that DCPIP can be used to measure the amount of ascorbic acid (vitamin C) present in fruit juice **J**. He found a bottle of solution labelled '1% ascorbic acid' in the school laboratory.

(i) Add 1.0 cm^3 of 1% ascorbic acid into a test-tube. Using a syringe, add DCPIP solution dropwise into the test-tube until in excess. Record your observations.

....

.... [1]

1. **Before** DCPIP was in excess, **drop of DCPIP** turn **colourless immediately**. **When DCPIP** was **in excess**, the mixture in test-tube turns **pink / yellow / blue**.

(ii) Use the information provided to plan a method that you can use to estimate the concentration of ascorbic acid in fruit juice **J**. Do **not** plan to carry out repeats.

You will carry out this method in **(c)(iii)**.

In your plan, you must use:
- 1% ascorbic acid
- DCPIP

You may select from any of the other apparatus and materials provided for **Q1**.

Your planned method should:

- have a clear and helpful structure so that the method described could be repeated by anyone reading it
- include details to ensure that results are as accurate and repeatable as possible
- use the correct technical and scientific terms
- only make use of the apparatus and materials provided.

<u>Reference to:</u>

<u>Variables + Procedures:</u>

1. **Independent variable**: <u>Concentration of ascorbic acid</u>
   - State the range (at least **5 values** at **fixed intervals**) for **ascorbic acid standards**.

2. Description on how to obtain or prepare the independent variable:
   - **Simple dilution** to get e.g. **1.0%, 0.8%, 0.6%, 0.4%, 0.2%** (the values must be stated). Full Description or dilution table to be given.

3. **Dependent variable**: <u>volume of DCPIP / number of drops of DCPIP needed to decolourise fixed volume of ascorbic acid</u>
   - Methodology on using **fixed volume of ascorbic acid** (state) and **titrate against variable amounts of DCPIP**;

4. **End point of titration** is **<u>colour indicated</u>** in **<u>(c) (i)</u>** where ascorbic acid no longer turns colourless when DCPIP is added;

5. Titrate fixed volume of unknown **J** with DCPIP and note the **volume / number of drops of DCPIP** needed;

6. **Compare** with **ascorbic acid standards** and **determine concentration** of unknown **J**;

[6]

(iii) Carry out your method described in **(c)(ii)** to estimate the concentration of ascorbic acid in fruit juice **J**.

(iv) Use the space below to record, in a suitable format, your results from **(c)(iii).**

Concentration of ascorbic acid / %	**Volume of DCPIP / ml <u>OR</u> number of drops of DCPIP (no units)**
1.0	Highest volume / drops
0.8	
0.6	
0.4	
0.2	Lowest volume
J	volume value closest to 0.6 & 0.4

[H]: Suitable headings with correct units; leftmost column for independent variable (concentration of ascorbic acid / %)

[O]: Correct and complete recording for volume of DCIP (correct to 1 decimal place) OR number of drops of DCPIP

[T]: Expected trend for volume or number of drops of DCIP

[3]

(v) State the concentration of the ascorbic acid in fruit juice **J**.

Accept: 0.4% – 0.6% (correct to 1 decimal place)

Note: to accept either a range or a fixed concentration

Concentration of ascorbic acid in fruit juice **J** …..…….% [1]

[Total: 27]

2 During this question you will require access to a microscope and slide **E1**.
An eyepiece graticule scale is attached to your microscope.

Slide **E1** is a microscope slide of a stained transverse section through a leaf of a land plant.
You are not expected to be familiar with this specimen.

(a) Observe slide **E1** using the 10X objective.

You are required to:

- use the eyepiece graticule to measure the depth of the midrib **Y** of the leaf as shown in Fig 2.1 and the depth of the vascular bundle at the midrib;
- draw a plan diagram of the leaf.

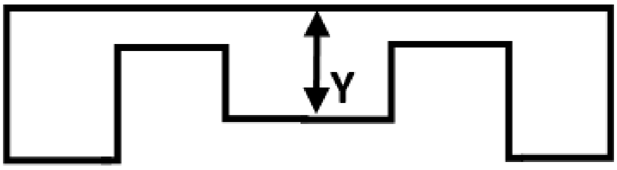


Fig 2.1

(i) Use the eyepiece graticule in the microscope to measure:

- the depth of the leaf at the mid-rib **Y**:

28 eyepiece graticule units

- the depth of the vascular bundle at the mid-rib :

7 eyepiece graticule units

Use these measurements to determine the simplest whole number ratio of the depth of the leaf at mid-rib **Y** to the depth of the vascular bundle at the mid-rib.

Show all the steps in your working.

Ratio of depth of leaf at midrib (Y) to depth of vascular bundle
= 28 / 7 or 28 : 7
= 4 / 1 or 4 : 1

[M] Correct measurement (in terms of eyepiece graticule (EPG) units) for depth of leaf at midrib and vascular bundle;

[C1] Correct calculation: number of EPG units of depth of leaf divided by depth of vascular bundle (appropriate statements are necessary) <u>and</u> Correct final answer in whole number and simplest form;

simplest whole number ratio 4 / 1 or 4 : 1

[2]

(ii) Outline how you can use the method described in **(a)(i)** to help you draw an accurate plan diagram of the leaf section in slide **E1** showing the arrangement and correct proportions (and shape) of the different tissues without calibrating the eyepiece graticule scale.

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……………………………………………………………………………………………… [2]

[How to measure the thickness of the different tissues]

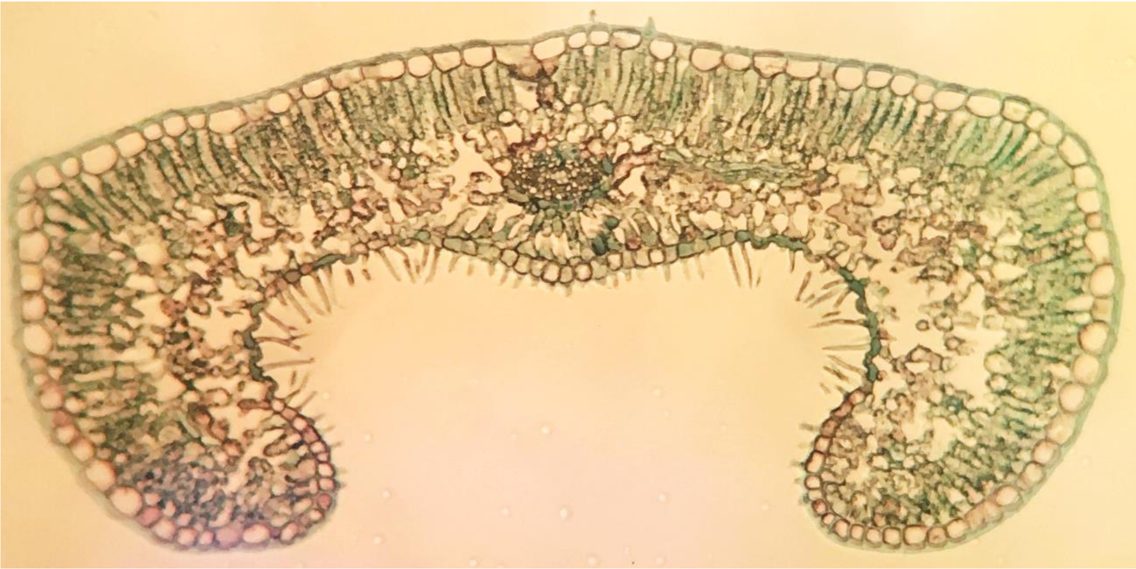
1. **Align** eyepiece graticule along the length and/or breath of the different tissues and measure their thickness by counting the **number of eyepiece graticule (EPG) units;**

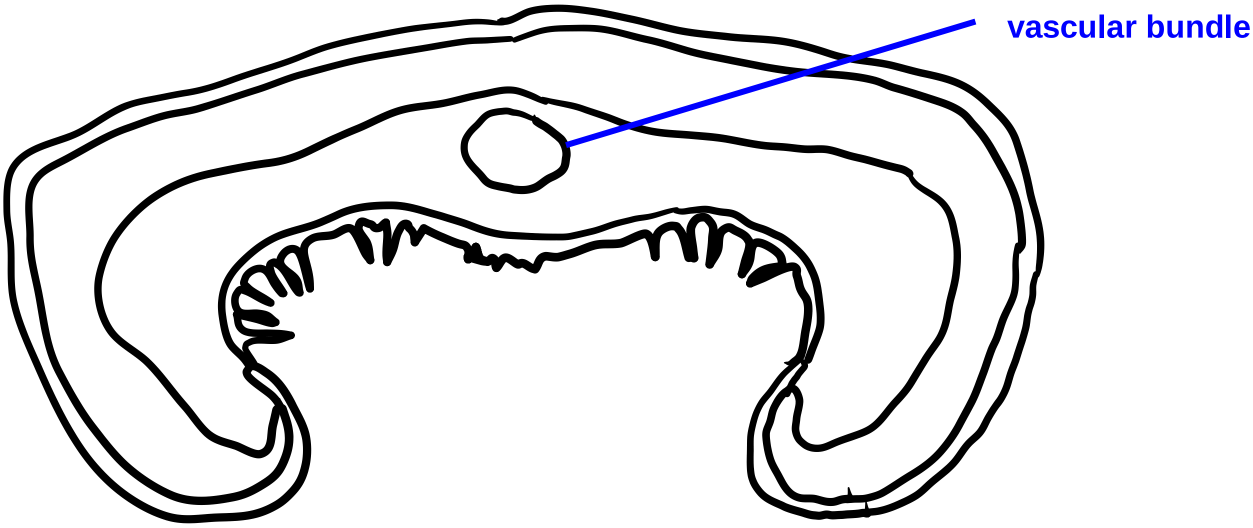
[How to determine proportion and represent in drawing]

2. Determine the *relative* proportion of each tissue by finding the ratio based on number of EPG units OR divide the length of each tissue by the thickness of the leaf section;
3. Use 10 mm to represent 20 EPG units in the drawing;

[2 max]

(iii) Draw a large plan diagram of the leaf section in slide **E1**. No cells should be drawn. Label one vascular bundle.





<u>**Plan drawing of Leaf, Slide E1 (T.S.) using 10X Objective**</u>

[4]

1. Size of plan drawing: more than half of given space or at least 100mm across widest dimension and quality drawing / smooth continuous lines;

   [Reject: shading; ruled lines for drawing; labels inside drawing]

   [Reject: any pencil line that is too thick or uneven; sketchy, feathery or dashed lines, gaps in line; overlapping lines; hanging lines]

2. Correct proportion and representation of tissue layers (no individual cells should be drawn in a plan diagram)
3. Flat top, midrib & correct shape;
4. Ruled/straight label line with label to "vascular bundle";

   [Reject: all other names]

(b) You are provided with a plastic ruler.

Use the plastic ruler to calibrate your eyepiece graticule under 100X magnification.

(i) Explain why this is a less accurate method compared to the use of a stage micrometer.

....

....

....

....

.... [2]

[Idea of alignment problem]

1. As the lines on the ruler are **thick**, there is difficulty in **aligning the eyepiece graticule** with the ruler for calibration as several eyepiece graticule units overlap

one marking on the ruler Or difficult to decide where to start and end the reading of the eyepiece graticule;

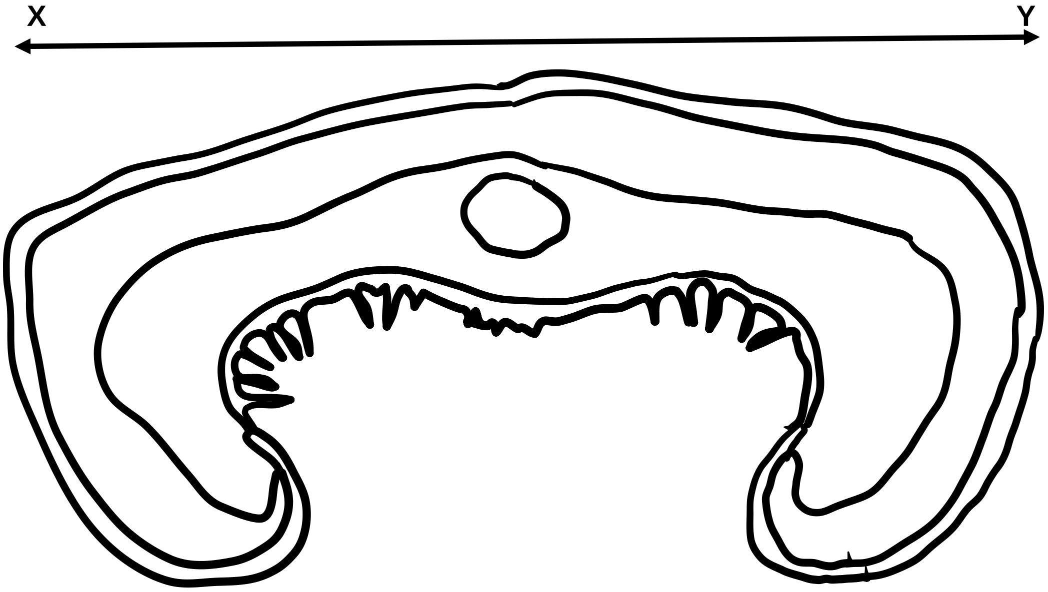
2. Compared to the stage micrometer which has thinner lines and the division superimpose accurately with the division on the eyepiece graticule;

[Uncertainty is larger with the ruler]

3. Smallest division on ruler is 1mm whereas that of stage micrometer is 0.01mm (note this will depend on the stage micrometer given);
4. Higher percentage error/ higher uncertainty with ruler;

(ii) Calculate the magnification of your plan drawing in **a(iii).**

Show your working clearly.



[2]

100 EPG divisions = 1 mm
1 EPG division = 1/100 mm

Actual length of XY = [100 + 26] EPG divisions
= 126 EPG divisions

$$\text{Magnification of Drawing} = \frac{\text{Drawing Length of (XY)}}{\text{Actual Length, XY}}$$

$$= \frac{116}{[126 \text{ X } 0.01]}$$

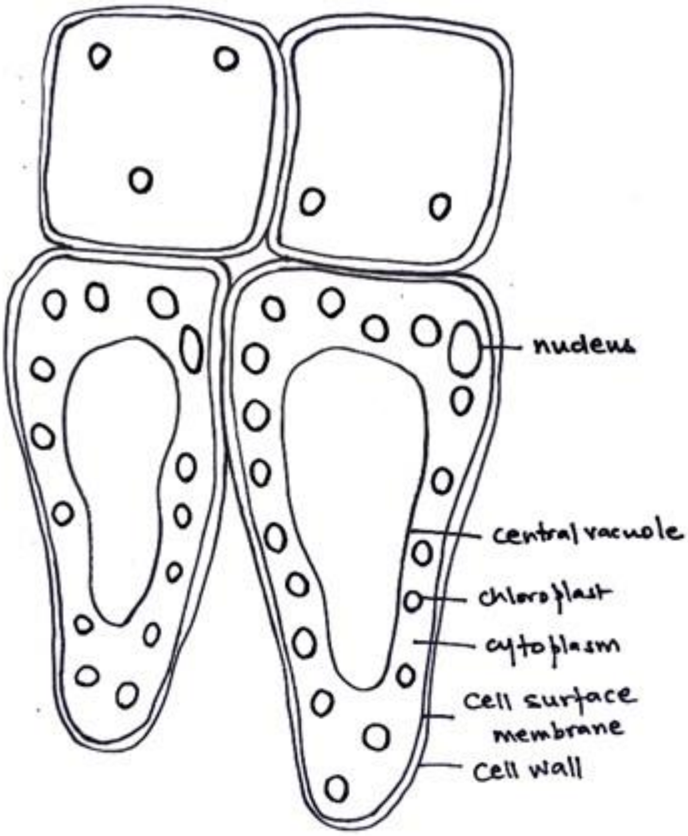
= 92 **X** or 92.1 **X** [Accept: Up to 1 dec. pl.]

100 eyepiece graticule (EPG) divisions = 1 mm on the ruler
[E/M] Magnification = Drawing Length, XY / Actual Length, XY
[C1] Correct calculation and Correct Final Answer expressed in whole number

[Note: drawing size must be indicated on the plan drawing, otherwise no marks)

(iii) Select a group of cells consisting of **two** adjacent palisade mesophyll cells that are touching **two** adjacent epidermal cells in slide **E1**.

Make a detailed drawing of this group of four cells and label any internal structures you can see in one of the palisade mesophyll cell.



[4]

[S] 2 palisade mesophyll & 2 epidermal cells drawn + size at least 50 mm along the widest dimension of the drawing;

[Q1] Use of sharp, continuous lines + no shading;

[Q2] Cells are drawn accurately; including

- cell walls drawn as double lines of appropriate thickness;
- several chloroplasts shown in palisade mesophyll cells;
- organelles are proportionate in size to cell size;
- no gaps between adjacent cells;

[L] Use of straight label lines to label at least **3 visible internal structures** (chloroplast / nucleus / cell wall / cell membrane / cytoplasm) of one palisade mesophyll cell

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(c) Fig 2.2 (on **page 17**) shows a photomicrograph of a transverse section of a leaf obtained from a plant that grows under a different environmental condition. Several structural differences are seen compared to that in slide **E1.**

Prepare the space below so that it is suitable for you to record these differences, other than size, between the leaves in slide **E1** and Fig. 2.2.

Record your observations in the space that you have prepared.

Any 3 of the following differences

	Feature	Leaf in Fig 2.2	Slide **E1**
1	Position of stomata	On upper epidermis	On lower epidermis
2	Number of stomata	numerous	Few *[Reject: none]*
3	Cuticle	Thinner cuticle / Absence of cuticle	Presence of think cuticle covering upper epidermis
4	Air spaces in spongy mesophyll *Reject: ref to air spaces in palisade mesophyll*	Numerous and Large	Few and small *[Reject: large airspaces absent]*
5a	Palisade mesophyll	Consists of numerous layers	Of 1-cell thick
5b	Proportion of palisade to spongy mesophyll	Thinner proportion	Equal proportion
6	Shape of leaf	<u>Flat</u> leaf blade	Blade <u>curled</u> downwards *[Reject: infolding / lobed]*
7	Trichomes (hair-like str) *[Reject: villi, follicles]*	Absent	Present
8	Epidermis thickness (relative to leaf)	Thinner	Thicker
9	Size / proportion of vascular bundle rel. to depth at midrib [Accept: correct ratios given]	Larger	Smaller

[4]

[1] Mark for appropriate table format, with lines (outside + inside) and one column for each specimen.

Reject: ref. to actual number of vascular bundles (different leaves from same plant type may differ slightly in number of VB)

Reject: comparisons of absolute measurements without mention of what these are relative to

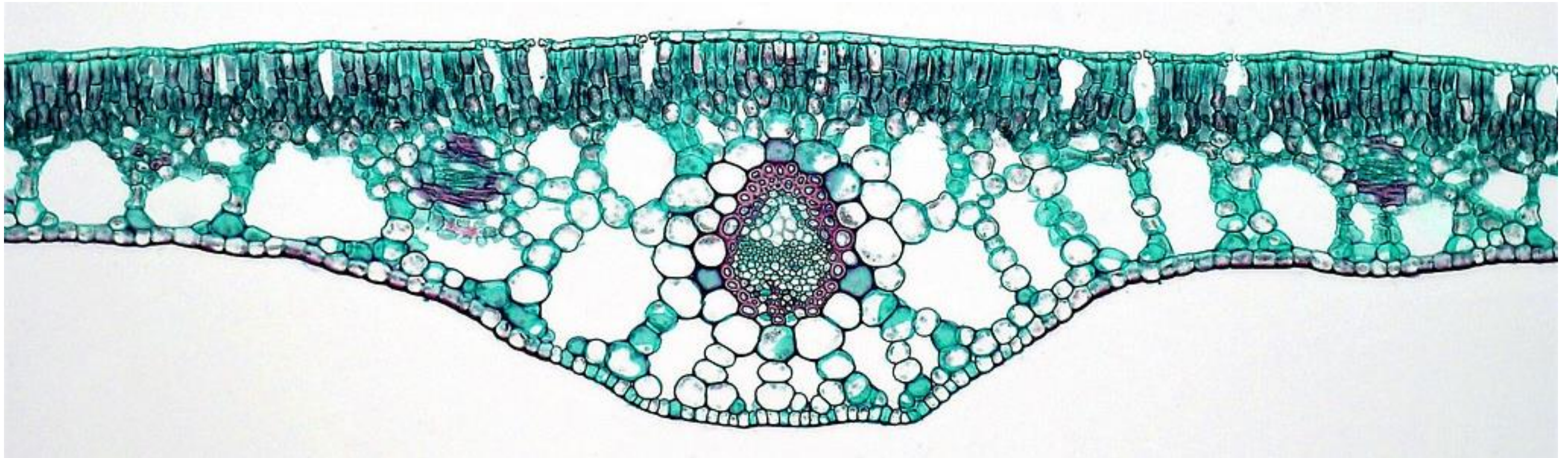

Fig 2.2

(d) "Slide **E1** is from a xerophytic plant that grows well in areas with low water availability."

State two visible features in the leaf specimen in slide **E1** that support the statement above.

....

....

....

.... [2]

Any 2 of the following:

1. Leaf is rolled inwards (to trap moisture in the leaf)
2. Hair-like structures (to trap moisture in the leaf) (**Reject: cilia, pili, root hairs)**
3. Few stomata (**Reject: "Few guard cells" without linkage to stomata)**
4. Stomata buried into pits (to prevent excess loss of water via transpiration)
5. Thick cuticle (to prevent water loss)
6. Thick epidermal cells/epidermis (to prevent water loss)
7. Reduced/small leaf size

(e) You are required to:

- find the area of the field of view, using Fig. 2.3 shown below;
- count the number of stomata in a quarter of the field of view, using Fig. 2.4 (on **page 19**);
- calculate the mean number of stomata per mm^2.

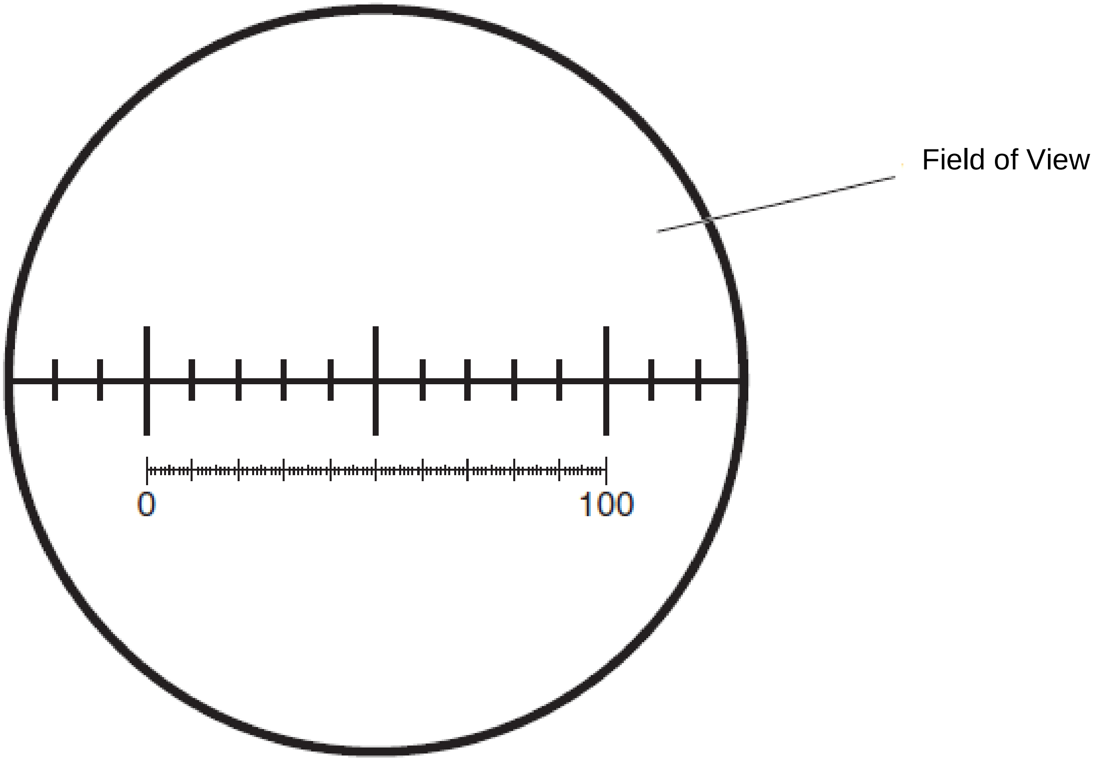

Fig 2.3

The smallest measurement on this stage micrometer is 0.1 mm.

(i) Calculate the area of the field of view (FOV), using the formula for the area of a circle: πr^2 where π = 3.14 and r = radius of the FOV.

Show your working and use appropriate units.

1. Diameter of field of view = 16 stage micrometer divisions
   = 16 x 0.1 = 1.6 mm

   Radius of field of view = 1.6 / 2
   = 0.8 mm

2. Area of field of view = 3.14 x 0.8^2 = 2.0096 = 2.01 mm^2 (up to 2 dec. pl.)

area of field of view (FOV)..................................... mm^2
[2]

Fig. 2.4 is a photomicrograph of the stomata in a field of view of the lower epidermis of a leaf from another plant. It was observed under the same magnification as in Fig. 2.3, hence under a field of view of the same size.

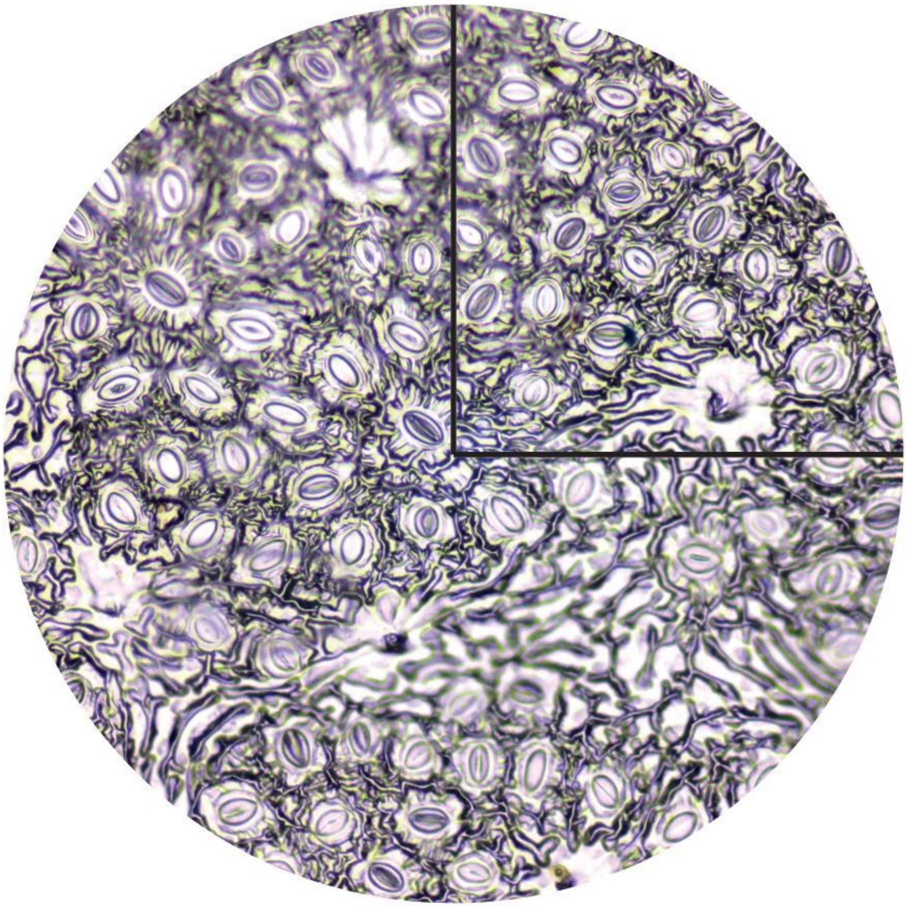

Fig. 2.4

There are too many stomata to count so the use of a sampling method is recommended to estimate the number of stomata in this field of view.

(ii) Count and record a sample number of **<u>whole</u>** stomata in the quarter of the field of view shown in Fig. 2.4.

Mark clearly with an '**X**' on each of the stomata counted in Fig. 2.4.

Number of stomata in the quarter of the field of view: ……….………...

[1]

Correctly marked with an "X" on the stomata counted within the quarter <u>and</u>

Correct number of stomata counted = 18 [Accept: 17 to 19]

(iii) Based on the methods used in **e(i)** and **e(ii)**, describe clearly the procedure used to estimate the mean number of stomata per mm^2 of this plant.

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……………………………………………………………………………………………………… [3]

1. Idea of using number of stomata counted in a quarter of the field of view (FOV) multiplied by 4 to obtain the estimate of the total number in the FOV
2. Idea of obtaining an average number of stomata per FOV from at least readings from 3 different sections / different leaves
3. Divide by the area of FOV (mm^2).

Accept: Average from at least 3 readings of number of stomata per FOV (mm^2)

[Total: 28]

END-OF-PAPER